

# SUBRAT KISHORE DUTTA

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## Education

### University of Hamburg

Doctorate in Philosophy

Hamburg, Germany

May 2025 – Present

- **Supervisors:** Dr. Xiao Zhang and Anne Lauscher
- **Research interest:** Adversarial Machine Learning, Fairness, Subpopulation

### Universität des Saarlandes

M.Sc. in Informatik | Grade: 1.2 / 1.0

Saarbrücken, Germany

April 2022 – Feb 2025

- **Thesis:** “Stealthy Targeted Adversarial Patch Attack through Perceptibility-Aware Optimization” | Grade: 1.0/1.0
- **Supervisors:** Dr. Xiao Zhang and Prof. Dr. Mario Fritz
- **Selected Courses:** Machine Learning, Image Processing and Computer Vision, Neural Network Theory and Implementation, Robustness in Machine Learning

### Assam Engineering College, ASTU

B.E. in Computer Science and Eng | First Class (Hons) Grade: 83.67% (Rank: 1)

Guwahati, Assam, India

Aug 2017 – Aug 2021

- **Thesis:** “Assamese Text Generation using Deep Learning Architectures”

## Publications and Preprints

**S. K. Dutta**, Y. Xu, P. Pant, and X. Zhang. (2025). *GREAT: Generalizable Backdoor Attacks in RLHF via Emotion-Aware Trigger Synthesis*. arXiv preprint arXiv:2510.09260. (**EMNLP 2026 Findings**)

**S. K. Dutta** and X. Zhang. (2025). *IAP: Invisible Adversarial Patch Attack through Perceptibility-Aware Localization and Perturbation Optimization*. In Proceedings of the IEEE/CVF International Conference on Computer Vision (**ICCV**).

Z. Chen, **S. K. Dutta**, Z. Zhao, C. Lin, C. Shen, and X. Zhang. (2024). *Can Targeted Clean-Label Poisoning Attacks Generalize?* arXiv preprint arXiv:2412.03908.

J. Patowary, **S. K. Dutta**, A. Mahanta, and T. T. Mantha. (2024). *A closer look at the air pollution dynamics during Diwali Festival in India*. **EGUsphere**.

**S. K. Dutta**, S. Saikia, A. Barman, R. Roy, K. Bora, L. B. Mahanta, and R. Suresh. (2021). *Study On Enhanced Deep Learning Approaches For Value-Added Identification And Segmentation Of Striation Marks In Bullets For Precise Firearm Classification*. *Applied Soft Computing*, Elsevier.

## Research Experience

### CISPA Helmholtz Center for Information Security

PhD Researcher

Saarbrücken, Germany

May 2025 – Present

- My research focuses on the intersection of adversarial machine learning and representational fairness of minority subpopulation in generative models and how the former can aid the latter.
- Supervisors: Dr. Xiao Zhang and Anne Lauscher

### CISPA Helmholtz Center for Information Security

Research Assistant

Saarbrücken, Germany

Jan 2024 – April 2025

- Conducting Visually Imperceptible Targeted Adversarial Patch Attacks through Perceptibility-Aware Optimization, under Dr. Xiao Zhang.
- Developed Perceptibility-Aware Optimization for patch localization.
- Proposed a novel patch update rule for colour constancy.
- The study was accepted at the International Conference on Computer Vision (ICCV) 2025.

### Max Planck Institute for Informatics

Research Assistant

Saarbrücken, Germany

Mar 2023 – Nov 2023

- D2: Computer Vision and Machine Learning.
- Studied performance improvements with generated samples using latent diffusion models in a few-shot learning setup under the guidance of Prof. Dr. Bernt Schiele and Dr. Anna Kukleva.

### Deutsches Forschungszentrum für Künstliche Intelligenz (SSE)

Research Assistant

Saarbrücken, Germany

August 2022 — Oct 2022

- Project Quasim.
- Developed the first and the second version of the Graphical User Interface for Project Quasim under the supervision of Wolfgang Maass. The GUIs were able to integrate all the machine learning-based functionalities. [Code]

### National Remote Sensing Centre, (ISRO)

Research Intern

Hyderabad, India

July 2021 — Oct 2021

- Studying the Long-term trend of water and vegetative indices in the Northern region of Alaska, under Dr. Santonu Goswami.
- Analyzing the correlation between the indices and the extent of the effect of one index over the others.

## Institute of Advanced Study in Science and Technology(IASST)

Guwahati, India

Research Intern

Jan 2020 – Oct 2020

- Analyzed the consistency of deep learning segmentation techniques for the task of segmentation of striation marks from fired bullets, under the guidance of Dr. Lipi B. Mahanta and the co-guidance of Dr. Kangkana Bora.
- The study was accepted in the journal of Applied Soft Computing, Elsevier.( doi.org/10.1016/j.asoc.2021.107789 )

## Technical Skills

- **Programming:** Python, C, C++
- **Libraries & Frameworks:** PyTorch, NumPy, Scikit-learn, Matplotlib, Hugging Face (Transformers, Diffusers), vLLM, TRL
- **Tools:** Docker, Git, Slurm, Weights & Biases (W&B), L<sup>A</sup>T<sub>E</sub>X

## Teaching and Mentoring Experience

- Co-advising bachelor's and master's theses on adversarial ML, including defenses against DreamBooth-style personalization and distributional backdoor poisoning
- Co-advised a completed master's thesis on robustness of defenses against diffusion-based facial customization.
- Teaching Assistant, University of Hamburg: Algorithmic Foundations of Adversarial Robustness (Spring 2025, 2026)

## Position of Responsibility

### Google Developer Student Clubs

Assam, India

Assam Engineering College

2020 – 2021

- Lead for Assam Engineering College (2020–2021).
- Created a community of more than 450 students and organized multiple technical sessions.
- Hosted the largest collaborative devfest in North-East India under the banner of DSC-EXPLORE.

### Google Cloud Facilitator

Assam, India

Assam Engineering College

2020 – 2021

- Facilitated the 30 Days of Google Cloud program in Assam Engineering College.

## Achievements

|                                                                                                     |       |
|-----------------------------------------------------------------------------------------------------|-------|
| <b>2021</b> Merit Overseas Research Scholarship Award 2021, Government of Assam                     | India |
| <b>2018, 2019, 2021</b> Merit Awards Fund for 3rd, 4th and 8th semesters, Assam Engineering College | India |
| <b>2020</b> Among Top 5 Teams, Smart India Hackathon (SIH)                                          | India |

## Languages

|                 |                          |
|-----------------|--------------------------|
| <b>English</b>  | Professional proficiency |
| <b>Hindi</b>    | Native proficiency       |
| <b>Assamese</b> | Native proficiency       |